

Codebook: Winter Weather

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Overview of data:

This codebook consists of data regarding meteorological winter events, with a focus on severe winter storm conditions and cold waves. The area in which the data comes from is Europe, which is then separated into four Subregions, Western Europe, Eastern Europe, Southern Europe, and Northern Europe. The data is from the years between 2000 and 2021 for each country where there was a recorded qualifying event. To determine the impact of winter weather per country, some variables are examined, including: the number of people affected and deaths, economic damage, and average temperatures. The way this data is set up allows for easy combination with other datasets (such as population to determine economic losses per capita, or governance scores to determine whether a stronger governance affects economic losses or total deaths, etc). Data is in descending order starting from the country with the most meteorological winter events.

Sources and Methodology

The data was sourced from the Centre for Research on the Epidemiology of Disasters' (CRED) and World Health Organization's Public Emergency Data (EM-DAT)

- EM-DAT, CRED / UCLouvain, 2025, Brussels, Belgium.

The data initially downloaded consisted of many more variables, such as specific starting and ending days and months, event name, reconstruction costs, insured damages, and other which pertained to larger datasets that would have information for said variables. The variables from the original data remain the same independent of the size or region from which the data is being sourced, which meant there was a lot of cleaning to do. At the end, I was able to clean the data to have only 10 variables while still maintaining all the important information.

I first removed the variables which were irrelevant to the data or that did not contain any values by using the "select(-###)" command. Then I grouped the data by countries, so that each row would refer to a specific country. In the process, I created a new variable called "Events" which indicates how many events each country faced within this time period. Then I also added total deaths, affected, and damage from each event per country, and calculated the mean deaths and damage as well. Finally, I arranged the data in descending order.

The next step was to rename the variables to comprehensible, self explanatory names without and under-scores. Then I had to further clean the data as there were still some missing information. I turned the 0's and NaN's into NA's by using the mutate function for consistency purposes.

Variable Presentation

1. ISO Country Codes

- Variable name: **rISO**
- Variable type: character
- Description: Standardized three-letter country codes that represented countries and dependent territories, as defined by the International Organization for Standardization.

2. Country

- Variable name: **rCountry**
- Variable type: character
- Description: Name of each country where a winter weather event occurred from 2000-2021.

3. Subregion

- Variable name: **rSubregion**
- Variable type: factor
- Description: Name of each subregion within Europe where a winter weather event occurred from 2000-2021.

Table 1: Frequency of Events by Subregion

Subregion	Count	Percent
Southern Europe	12	36.4%
Eastern Europe	10	30.3%
Western Europe	6	18.2%
Northern Europe	5	15.2%

4. Number of Winter Weather Events per Country

- Variable name: **rEvents**
- Variable type: numeric
- Description: Number of winter weather events per country triggered by “Cold wave” or “Severe winter conditions” as determined by EM-DAT from 2000-2021. “Cold wave” is defined as “A period of abnormally cold weather”, and “Severe winter conditions” is defined as “Damage caused by snow and ice. Winter damage refers to damage to buildings, infrastructure, traffic (esp. navigation) inflicted by snow and ice in the form of snow pressure, freezing rain, frozen waterways etc.”

Table 2: Statistics for Number of Events

Min	Mean	Median	Max
1	3	2	6

5. Total Deaths

- Variable name: **rDeaths**
- Variable type: numeric
- Description: Aggregated number of deaths from all winter weather events per country from 2000-2021.

Table 3: Statistics for Total Deaths

Min	Mean	Median	Max
1	182.8387	23	1817

6. Average Deaths

- Variable name: **rAvg_deaths**
- Variable type: numeric
- Description: Average number of deaths from all winter weather events per country from 2000-2021.

Table 4: Statistics for Average Deaths

Min	Mean	Median	Max
1	182.8387	23	1817

7. Total Affected

- Variable name: **rAffected**
- Variable type: numeric
- Description: Total number of people affected by winter weather events per country from 2000-2021.

Table 5: Statistics for Total People Affected

Min	Mean	Median	Max
20	44392.94	14477	230000

8. Total Damage (thousands, USD)

- Variable name: **rDamage**
- Variable type: numeric
- Description: The total damage in thousands reported in USD from winter weather events per country from 2000-2021.

Table 6: Statistics for Total Damage (thousands, USD)

Min	Mean	Median	Max
53615	1258981	175979	6297123

9. Average Damage

- Variable name: **rAvg_damage**
- Variable type: numeric
- Description: The average damage in thousands reported in USD from all winter weather events per country from 2000-2021.

Table 7: Statistics for Average Damage (thousands, USD)

Min	Mean	Median	Max
53615	1258981	175979	6297123

10. Average Temperature

- Variable name: `rAvg_temp`
- Variable type: numeric
- Description: Aggregated average temperature recorded during each winter weather event in Celsius per country from 2000-2021.

Table 8: Statistics for Average Temperature

Min	Mean	Median	Max
-37.25	-26.68144	-26.76667	-11

```
knitr::opts_chunk$set(echo = FALSE,
  eval = TRUE,
  fig.align = 'center',
  message = FALSE,
  warning = FALSE
)

library(janitor)
library(dplyr)
library(readxl)
library(scales)
library(knitr)
library(tinytex)
library(tidyrr)
cold <- read_excel("Extreme Cold.xlsx")

cold <- cold %>%
  select(ISO,
    Country,
    Subregion,
    Magnitude,
    `Magnitude Scale`,
    `Start Year`,
    `Total Deaths`,
    `Total Affected`,
    `Total Damage, Adjusted ('000 US$)`)

cold <- cold %>%
  rename(
    Year = `Start Year`,
    Measure = `Magnitude Scale`,
    Deaths = `Total Deaths`,
    Affected = `Total Affected`,
    Damage = `Total Damage, Adjusted ('000 US$)`
```

```

)

bycountry <- cold %>%
  group_by(ISO, Country, Subregion) %>%
  summarize(
    Events      = n(),                                # count of events
    Deaths     = sum(Deaths, na.rm = TRUE),          # sum of deaths
    Avg_deaths  = mean(Deaths, na.rm = TRUE),         # average deaths per event
    Affected    = sum(Affected, na.rm = TRUE),        # sum of people affected
    Damage      = sum(Damage, na.rm = TRUE),
    Avg_damage  = mean(Damage, na.rm = TRUE),
    Avg_temp    = mean(Magnitude, na.rm = TRUE),
  ) %>%
  arrange(desc(Events))

bycountry <- bycountry %>%
  mutate(across(
    where(is.numeric),
    ~ na_if(., 0)
  ))

bycountry <- bycountry %>%
  mutate(across(
    where(is.numeric),
    ~ replace(., is.nan(.), NA)
  ))

subregionfreq <- bycountry %>%
  ungroup() %>%
  count(Subregion, name = "frequency") %>%
  arrange(desc(frequency))

subregionfreq <- subregionfreq %>%
  mutate(percent = scales::percent(frequency / sum(frequency)))

subregionfreq %>%
  kable(
    col.names = c("Subregion", "Count", "Percent"),
    caption   = "Frequency of Events by Subregion"
  )

eventfreq <- bycountry %>%
  ungroup() %>%
  count(Events, name = "evCount")

eventsummary <- eventfreq %>%
  summarize(
    Min    = min(evCount),
    Mean   = mean(evCount),
    Median = median(evCount),
    Max    = max(evCount)
  )

eventsummary %>%

```

```

kable(
  col.names = c("Min", "Mean", "Median", "Max"),
  caption   = "Statistics for Number of Events"
)
deathstats <- bycountry %>%
  ungroup() %>%
  summarize(
    Min    = min(Deaths,    na.rm = TRUE),
    Mean    = mean(Deaths,   na.rm = TRUE),
    Median  = median(Deaths, na.rm = TRUE),
    Max     = max(Deaths,    na.rm = TRUE)
  )

deathstats %>%
  kable(
    col.names = c("Min", "Mean", "Median", "Max"),
    caption   = "Statistics for Total Deaths"
  )
avgdeathstats <- bycountry %>%
  ungroup() %>%
  summarize(
    Min    = min(Avg_deaths, na.rm = TRUE),
    Mean    = mean(Avg_deaths, na.rm = TRUE),
    Median  = median(Avg_deaths, na.rm = TRUE),
    Max     = max(Avg_deaths, na.rm = TRUE)
  )

avgdeathstats %>%
  kable(
    col.names = c("Min", "Mean", "Median", "Max"),
    caption   = "Statistics for Average Deaths"
  )
affectedstats <- bycountry %>%
  ungroup() %>%
  summarize(
    Min    = min(Affected, na.rm = TRUE),
    Mean    = mean(Affected, na.rm = TRUE),
    Median  = median(Affected, na.rm = TRUE),
    Max     = max(Affected, na.rm = TRUE)
  )

affectedstats %>%
  kable(
    col.names = c("Min", "Mean", "Median", "Max"),
    caption   = "Statistics for Total People Affected"
  )
damagestats <- bycountry %>%
  ungroup() %>%
  summarize(
    Min    = min(Damage, na.rm = TRUE),
    Mean    = mean(Damage, na.rm = TRUE),
    Median  = median(Damage, na.rm = TRUE),
    Max     = max(Damage, na.rm = TRUE)
  )

```

```

)

damagestats %>%
  kable(
    col.names = c("Min", "Mean", "Median", "Max"),
    caption = "Statistics for Total Damage (thousands, USD)"
  )
avgdamagestats <- bycountry %>%
  ungroup() %>%
  summarize(
    Min = min(Avg_damage, na.rm = TRUE),
    Mean = mean(Avg_damage, na.rm = TRUE),
    Median = median(Avg_damage, na.rm = TRUE),
    Max = max(Avg_damage, na.rm = TRUE)
  )

avgdamagestats %>%
  kable(
    col.names = c("Min", "Mean", "Median", "Max"),
    caption = "Statistics for Average Damage (thousands, USD)"
  )
temperature <- bycountry %>%
  ungroup() %>%
  summarize(
    Min = min(Avg_temp, na.rm = TRUE),
    Mean = mean(Avg_temp, na.rm = TRUE),
    Median = median(Avg_temp, na.rm = TRUE),
    Max = max(Avg_temp, na.rm = TRUE)
  )

temperature %>%
  kable(
    col.names = c("Min", "Mean", "Median", "Max"),
    caption = "Statistics for Average Temperature"
  )

```